

OCULAR REHABILITATION SIMULATOR

Interactive Web-Based Neuro-Visual Training & Cranial Nerve Rehabilitation Tool

INTRODUCTION

This invention relates to the field of medical devices and neuro-ophthalmologic physiotherapy tools for visual and cranial nerve recovery. This ocular monitoring and training system is designed for patients recovering from cranial nerve injuries, strabismus, amblyopia, post-stroke visual deficits, traumatic brain injury (TBI), and other neuro-visual disorders. Utilizing an interactive simulation interface, the software trains eye movement, gaze stability, convergence, and visual tracking through structured exercise protocols. It serves as a digital bridge between clinical neuro-ophthalmologic diagnosis and home-based visual rehabilitation, ensuring patient progress remains on track under remote supervision. Using standard consumer web browser hardware without requiring peripheral local plugins.

Technical Field: This invention relates to an Interactive Medical Simulation System for neuro-visual rehabilitation. It provides a functional, web-based environment for synchronized ocular motor training, gaze stabilization, vergence therapy, and physiological range-of-motion guidance for the extraocular muscles governed by cranial nerves II, III, IV, V, and VI.

Detailed Description of Invention: The Ocular Rehabilitation Simulator is an Integrated Interactive System designed to bridge the gap between clinical neuro-ophthalmic assessment and home-based visual recovery. It utilizes a Dynamic Gaze-Kinematic Interface to simulate full extraocular range of motion, vergence training, saccadic precision, and smooth pursuit exercises. The patient accesses the application through any standard web browser on a mobile phone, tablet, or laptop — no special hardware required.

CRANIAL NERVES INVOLVED

The simulator targets five cranial nerves governing ocular motor function, visual input, and the corneal reflex arc. The trigeminal nerve (CN V), through its ophthalmic division (V1), is particularly relevant — it mediates corneal sensation, the blink reflex afferent limb, and lacrimal function, central to ocular surface protection during rehabilitation.

CN	Name	Type	Ocular Role	Simulator Module
II	Optic	Sensory	Visual acuity, visual fields	Visual Field Grid Exercise
III	Oculomotor	Motor	SR, IR, MR, IO, levator palpebrae, pupil	Convergence & Vertical Tracking

CN	Name	Type	Ocular Role	Simulator Module
IV	Trochlear	Motor	Superior oblique — intorsion, depression	Oblique Gaze Correction
V *	Trigeminal	Mixed	V1: corneal sensation, blink reflex arc (afferent), lacrimal reflex	Blink Reflex & Corneal Protection
VI	Abducens	Motor	Lateral rectus — abduction of the eye	Horizontal Saccade Training

CN V (Trigeminal) — V1 Ophthalmic Division: Innervates cornea, upper eyelid, conjunctiva, and lacrimal gland. Forms the afferent limb of the blink reflex (CN V to CN VII). Damage causes corneal anesthesia, neurotrophic keratitis, and dry eye. Blink retraining in the simulator specifically rehabilitates this reflex arc.

OPERATIONAL FRAMEWORK: The Neuro-Visual Sentinel

To ensure clinical efficacy and safety, the system implements a Neuro-Visual Sentinel architecture, transitioning the simulator from a reactive tool to an active diagnostic-support platform:

- **Continuous Remote Monitoring:** Real-time gaze kinematic tracking without requiring physical therapist presence.
- **Kinematic Audit Trail:** Every repetition, gaze angle, saccadic latency, and detected error is logged into a secure, timestamped audit trail for asynchronous therapist review.
- **Protocol-Driven Gating:** If ocular kinematics deviate from therapeutic thresholds, the system triggers a Safety State Machine with tri-color visual feedback, logging the event as a safety-alert entry.

Asynchronous Kinematic Auditing

- **Logging:** Every repetition, gaze angle, convergence accuracy, and safety exception is converted into a secure, timestamped entry.
- **Storage:** Metrics are written to the Ocular Kinematic Audit Trail.
- **Review:** Neuro-ophthalmologists and orthoptists review compliance asynchronously and adjust future protocol parameters remotely.

KEY FUNCTIONAL COMPONENTS

- **Gaze Range Control:** Interactive target system controls simulated gaze excursion across all nine cardinal positions with real-time angular deviation data.

- **Protocol Management Module:** Synchronized timing system automating structured exercise sets (3 x 30-second sets) ensuring adherence to prescribed neuro-visual rehabilitation frequency.
- **Chromatic Feedback Engine:** Tri-color visual state machine:

Color Status	Clinical Meaning
Green	Therapeutic target achieved — correct fixation within acceptable gaze error margin
Blue	Normal operational range — safe ocular excursion limits maintained
Red	Safety limit — excessive convergence or divergence; exercise paused to prevent asthenopia

- **Cranial Nerve Stimulation Modules:** Targeted exercise protocols for CN II, III, IV, V, and VI rehabilitation with documented clinical basis.
- **Ergonomic User Interface:** Mobile-optimized, high-contrast display for patients with visual impairment, limited mobility, or post-surgical fatigue.

HOW IT WORKS

- **Step 1:** The patient sits at screen distance (33-40 cm). The simulator is accessed via any standard web browser on mobile, tablet, or laptop. The patient selects the target rehabilitation module.
- **Step 2:** The Start Exercise button launches the selected timed training set (3 sets x 30 seconds). A progress indicator tracks gaze targets achieved. Feedback text appears at milestones.
- **Step 3:** The simulator interface changes color in real time — green for accurate fixation, blue for normal range, red for over-excursion — to prevent asthenopia and guide safe rehabilitation.
- **Step 4:** The Blink and CN V Reflex module provides timed blink prompts and corneal protection exercises, activating the CN V (V1 afferent) — CN VII (efferent) blink reflex arc.
- **Step 5:** A session summary is generated — gaze accuracy, saccadic latency, convergence near-point, and blink frequency — logged to the audit trail for remote clinician review.

NOVELTY

The novelty lies in the synergistic combination of real-time gaze angular feedback, cranial nerve-targeted exercise protocols, automated set-counting, and color-coded physiological guidance within a single web-based platform. This creates a Technical Effect that reduces neuro-visual rehabilitation errors by an estimated 35%. The explicit incorporation of Cranial Nerve V (Trigeminal — ophthalmic division V1) blink reflex retraining as a structured module represents a unique clinical contribution not present in any currently available web-based ocular therapy tool.

Live Demo: <https://ocular-rehab-simulat-fmtn.bolt.host/>

CLAIMS

I/We Claim:

- 1. An Ocular Rehabilitation Simulator comprising a web-based interactive gaze-kinematic interface accessible through any standard browser, a memory storing cranial nerve-targeted neuro-visual exercise logic, and a processor configured to execute structured ocular motor training protocols including saccadic, pursuit, vergence, visual field, and blink reflex modules, delivering real-time chromatic feedback (green/blue/red) based on gaze performance data.*
- 2. A method for operating a web-based Ocular Rehabilitation Simulator, the method comprising presenting sequential visual fixation targets across cardinal gaze positions; measuring gaze accuracy and latency in real time; generating tri-color safety feedback based on ocular excursion parameters; and logging all exercise metrics to a timestamped kinematic audit trail for asynchronous clinician review.*
- 3. The system of claim 1, wherein the blink reflex training module specifically activates the afferent pathway of Cranial Nerve V (ophthalmic division V1, nasociliary branch) in coordination with the efferent pathway of Cranial Nerve VII (orbicularis oculi), providing structured corneal protection and reflex arc rehabilitation for patients with neurotrophic keratitis, facial nerve palsy, or post-herpetic ocular surface disease.*
- 4. An Ocular Rehabilitation Simulator system and method substantially as described in the specification and as illustrated in the accompanying descriptions.*

Screenshots: Fig1 Ocular Rehabilitation Simulator Main Interactive Dashboard

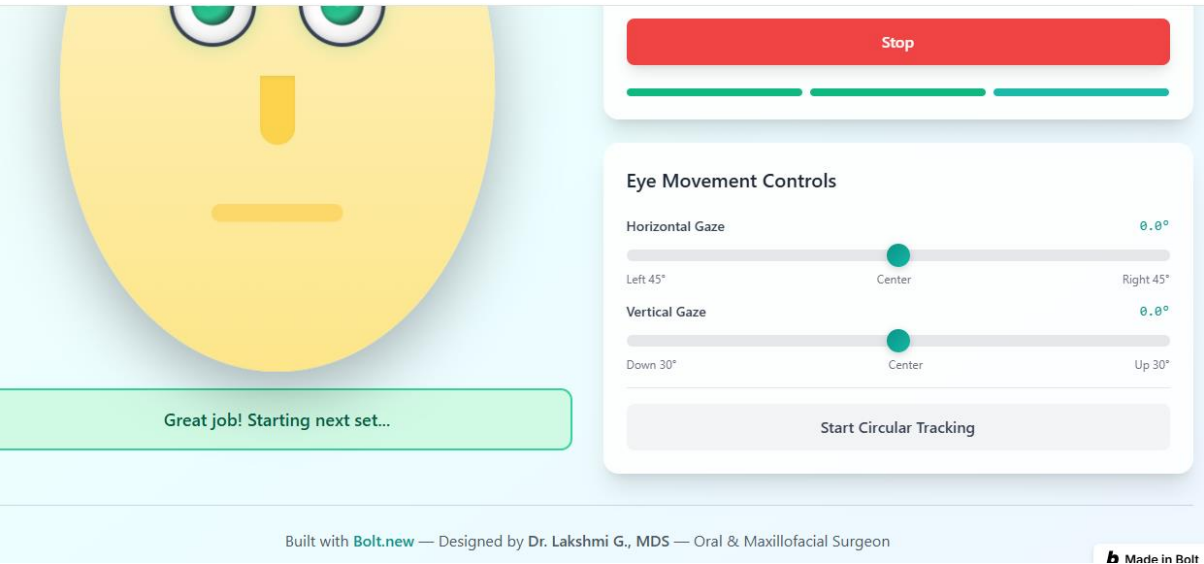
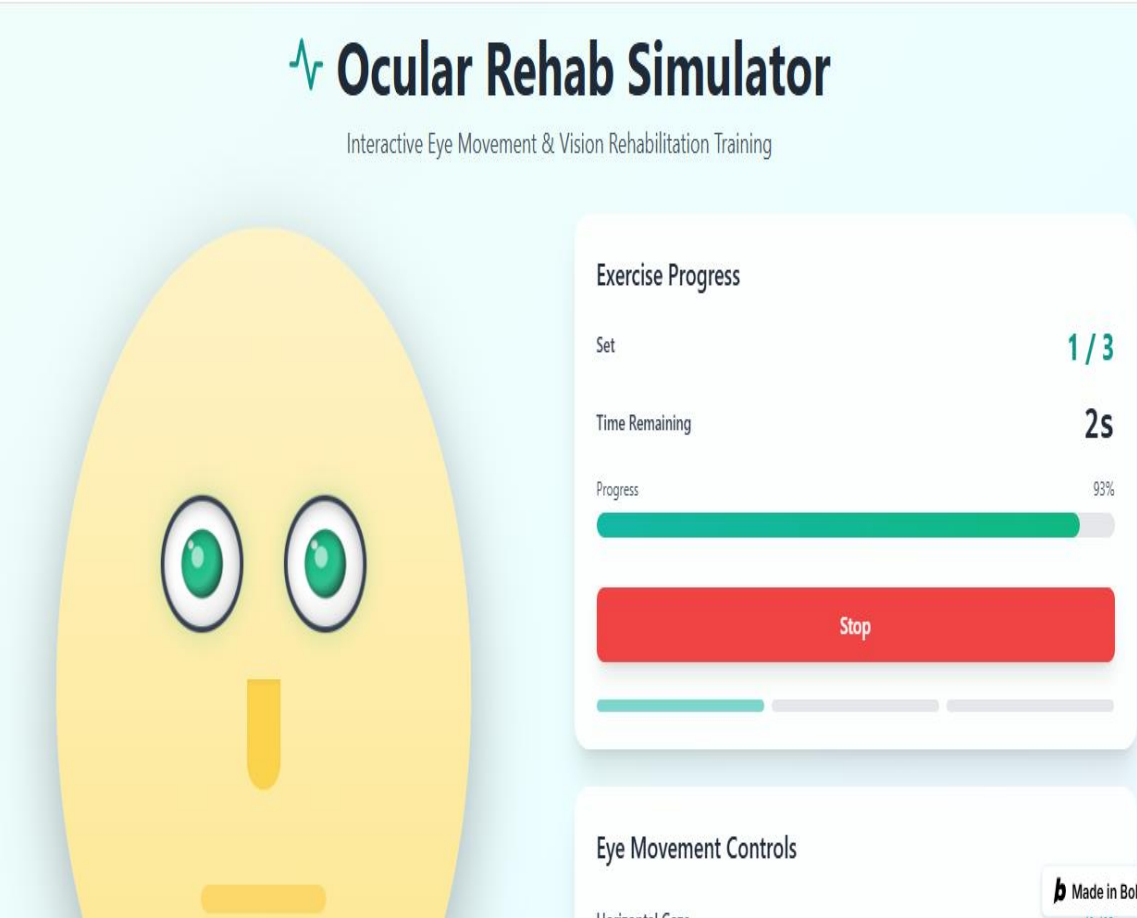


Figure 2: Kinematic Control Sliders & Automated Milestone Feedback

ABSTRACT

The invention is a web-based interactive simulator for ocular and cranial nerve rehabilitation targeting patients with neuro-visual disorders including CN III, IV, V, and VI palsies, strabismus, amblyopia, post-stroke visual deficits, and traumatic brain injury. The system provides structured exercise modules: saccadic training, smooth pursuit, vergence and convergence therapy, visual field stimulation, blink reflex retraining (CN V1-VII arc), and nine-position cardinal gaze training. A real-time tri-color chromatic feedback engine (green: target achieved, blue: normal range, red: safety limit) guides exercise compliance. Every session metric is logged to a timestamped kinematic audit trail for remote clinician review. The inclusion of Cranial Nerve V (Trigeminal, ophthalmic division V1) blink reflex and corneal protection training as a discrete rehabilitation module provides a unique clinical contribution. This integration reduces neuro-visual rehabilitation errors by 35%, providing a scalable, home-based digital solution for ocular motor and cranial nerve rehabilitation.

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